



6970 - The diversity of the progenitors of Fast X-ray Transients

Cycle: 3, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1		NIRCam Imaging	(7) EP250306a
	2		NIRSpec IFU Spectroscopy	(4) FXT1-trigger
	4		NIRSpec IFU Spectroscopy	(4) FXT1-trigger
	3		NIRCam Imaging	(7) EP250306a
	5		NIRCam Imaging	(7) EP250306a
	6		NIRSpec IFU Spectroscopy	(6) EP250207b

ABSTRACT

Fast X-ray transients (FXTs) are extra-galactic flashes of X-rays. Einstein Probe (EP) carries the promise to revolutionise the field, quickly ($\ll 1$ d) informing us on each of the many FXTs it discovers (>50 in ~ 0.5 year). However, obtaining the diagnostics required to determine their progenitors has been hard. The 2 best-studied FXTs include a peculiar $z \sim 5$ long gamma-ray burst and a relativistic supernova with properties linking it to luminous fast blue optical transients. The question is if these are two examples of the extreme ends of the luminosity function of the same population or is the new EP detection method allowing us to extend the range of progenitors that we probe? To understand this we urgently need new the diagnostic power that XMM-Newton, VLT/MUSE and JWST can provide.

OBSERVING DESCRIPTION

We propose to obtain either NIRSpec/IFU or NIRCAM imaging of any accompanying supernova, and likely the host galaxy of 3 newly discovered Fast X-ray

Transients. The first observation should take place ~ 21 days after triggering. The second 0.5-1 year later.

For NIRCAM we will observe in 4 filters (F090W, F115W, F277W, F356W,), to ensure that afterglow, supernova and host are all well characterised and distinguishable. In each case we will obtain 1245 s exposure. This will yield 3-sigma AB mag detection limits of about F090W=28.4, F115W=28.5, F277W=29.0, F356W=29.2.

Proposal 6970 - Targets - The diversity of the progenitors of Fast X-ray Transients

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(4)	FXT1-trigger	RA: 19 19 32.5578 (289.8856575d) Dec: -43 22 52.17 (-43.38116d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr	
	Comments: Category=Unidentified Description=[X-ray sources]				
	(5)	FXT1-acquisition	RA: 19 19 32.9120 (289.8871333d) Dec: -43 23 8.63 (-43.38573d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr	
	Comments: Category=Unidentified Description=[X-ray sources]				
Fixed Targets	(6)	EP250207b	RA: 11 10 3.2069 (167.5133621d) Dec: -07 52 7.34 (-7.86871d) Equinox: J2000	Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Epoch of Position: 2000	
	Comments: Category=Unidentified Description=[X-ray sources] Extended=YES				
	(7)	EP250306a	RA: 09 43 47.7600 (145.9490000d) Dec: +15 41 37.32 (15.69370d) Equinox: J2000	Epoch of Position: 2000	
	Comments: Category=Unidentified Description=[X-ray sources] Extended=NO				
Generic Targets	#	Name	Criteria	Description	
	(1)	FXT1	Einstein Probe discovered Fast X-ray Transient 1		
	Comments: 2 observations either imaging with NIRCAM or NIRSpec IFU				
	(2)	FXT2	Einstein Probe discovered Fast X-ray Transient 2		
Generic Targets	Comments: 2 observations either imaging with NIRCAM or NIRSpec IFU				
	(3)	FXT3	Einstein Probe discovered Fast X-ray Transient 3		
Comments: 2 observations either imaging with NIRCAM or NIRSpec IFU					

Proposal 6970 - Observation 1 - The diversity of the progenitors of Fast X-ray Transients

Observation	Proposal 6970, Observation 1 Tue May 26 19:00:13 GMT 2026																																							
	Diagnostic Status: Warning Observing Template: NIRCcam Imaging <i>Comments: For FXT1 epoch 1 and 2 either this or Observation 2 will be executed, not both. These are observing templates! The epoch 2 (the second observation) should be done 0.5-1 yr after the first.</i>																																							
Diagnosics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Observation 1) Informational (Form): Minimum Response time is a required field.																																							
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Special Requirements	Before Date 10-APR-2026 Target Of Opportunity Response Time <None specified> to 21 Days, Carry-Over Fiducial Point Override NRCB1_FULLL																																							

Proposal 6970 - Observation 2 - The diversity of the progenitors of Fast X-ray Transients

Observation	Proposal 6970, Observation 2											Tue May 26 19:00:13 GMT 2026
	Diagnostic Status: Warning											
Observing Template: NIRSpec IFU Spectroscopy												
Diagnostics	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Observation 2) Informational (Form): Minimum Response time is a required field.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	FXT1-trigger	RA: 19 19 32.5578 (289.8856575d) Dec: -43 22 52.17 (-43.38116d) Equinox: J2000				Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr					
	Comments: Category=Unidentified Description=[X-ray sources]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	5 FXT1-acquisition	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	264145	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	41	1	false	true	WAVECAL	4	4	2450.934	223693

Proposal 6970 - Observation 2 - The diversity of the progenitors of Fast X-ray Transients

Special Requirements	No Parallel Attachments Target Of Opportunity Response Time <None specified> to 21 Days, Carry-Over
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Proposal 6970 - Observation 4 - The diversity of the progenitors of Fast X-ray Transients

Observation	Proposal 6970, Observation 4											Tue May 26 19:00:13 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
	(Observation 4) Informational (Form): Minimum Response time is a required field.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(4)	FXT1-trigger	RA: 19 19 32.5578 (289.8856575d)				Proper Motion RA: 0 mas/yr					
			Dec: -43 22 52.17 (-43.38116d)				Proper Motion Dec: 0 mas/yr					
			Equinox: J2000									
	Comments: Category=Unidentified Description=[X-ray sources]											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	5 FXT1-acquisition	WATA	SUB2048	F110W	NRSRAPID	3	1	1	3.628	264145	
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type			Size	Starting Point			Number of Points		Points	
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	43	1	false	true	NONE	4	4	2567.645	223693

Proposal 6970 - Observation 4 - The diversity of the progenitors of Fast X-ray Transients

Special Requirements	Between Dates 03-APR-2026:00:00:00 and 22-MAY-2026:00:00:00 Target Of Opportunity Response Time <None specified> to 21 Days, Carry-Over
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Proposal 6970 - Observation 3 - The diversity of the progenitors of Fast X-ray Transients

Observation	Proposal 6970, Observation 3 Tue May 26 19:00:13 GMT 2026																																							
	Diagnostic Status: Warning Observing Template: NIRCam Imaging <i>Comments: For FXT2 epoch 1 and 2 either this or Observation 4 will be executed, not both. These are observing templates!</i> <i>The epoch 2 (the second observation) should be done 0.5-1 yr after the first.</i>																																							
Diagnosics	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Observation 3) Informational (Form): Minimum Response time is a required field.																																							
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2	F115W	F356W	SHALLOW4	6	1	4	4	1245.465	224262																															
Special Requirements	Between Dates 10-MAY-2026 and 01-JUN-2026 Target Of Opportunity Response Time <None specified> to 21 Days, Carry-Over Fiducial Point Override NRCB1_FULL																																							

Proposal 6970 - Observation 5 - The diversity of the progenitors of Fast X-ray Transients

Observation	Proposal 6970, Observation 5										Tue May 26 19:00:13 GMT 2026
	Diagnostic Status: Warning										
	Observing Template: NIRCcam Imaging										
	Comments: For FXT3 epoch 1 and 2 either this or Observation 6 will be executed, not both. These are observing templates! The epoch 2 (the second observation) should be done 0.5-1 yr after the first.										
Diagnostics	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name	Target Coordinates			Targ. Coord. Corrections			Miscellaneous		
	(7)	EP250306a	RA: 09 43 47.7600 (145.9490000d) Dec: +15 41 37.32 (15.69370d) Equinox: J2000			Epoch of Position: 2000					
	Comments: Category=Unidentified Description=[X-ray sources] Extended=NO										
Template	Module		Subarray				Target Placement				
	ALL		FULL				Detector B1 (compact source)				
Dithers	#	Primary Dither Type		Primary Dithers		Subpixel Dither Type		Dither Size		Subpixel Positions	
	1	INTRAMODULEX		4		STANDARD				1	
Spectral Elements	#	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F150W	F444W	SHALLOW4	6	1	4	4	1245.465		
	2	F115W	F356W	SHALLOW4	6	1	4	4	1245.465		
Special Requirements	After Date 10-NOV-2026:00:00:00 Target Of Opportunity Response Time 14 Days to 21 Days, Carry-Over Fiducial Point Override NRCB1_FULL										

Proposal 6970 - Observation 6 - The diversity of the progenitors of Fast X-ray Transients

Observation	Proposal 6970, Observation 6 Tue May 26 19:00:13 GMT 2026											
	Diagnostic Status: Warning Observing Template: NIRSpec IFU Spectroscopy <i>Comments: For FXT3 epoch 1 and 2 either this or Observation 5 will be executed, not both. These are observing templates!</i> <i>The epoch 2 (the second observation) should be done 0.5-1 yr after the first.</i>											
Diagnosics	(Visit 6:1) Warning (Form): Overheads are provisional until the Visit Planner has been run. (Observation 6) Informational (Form): Minimum Response time is a required field.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(6)	EP250207b	RA: 11 10 3.2069 (167.5133621d) Dec: -07 52 7.34 (-7.86871d) Equinox: J2000				Proper Motion RA: 0 mas/yr Proper Motion Dec: 0 mas/yr Epoch of Position: 2000					
Template	<i>Comments:</i> Category=Unidentified Description=[X-ray sources] Extended=YES											
	TA Method						HFF Readout Mode					
	NONE						false					
Dithers	#	Dither Type		Size		Starting Point		Number of Points		Points		
	1	4-POINT-DITHER										
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRSIRS2RAPID	54	1	false	true	NONE	4	4	3209.556	
Special Requirements	Target Of Opportunity Response Time <None specified> to 21 Days, Carry-Over											